

Practical Django Project Question

Project: Student Course Management System

Build a Django web application where users can register, log in, and manage courses. The system should support different user roles using a **custom user model**.

Requirements

1 Custom User Model

Create a custom user model by extending:

- AbstractUser or AbstractBaseUser

Add the following additional fields:

- role (choices: Admin, Instructor, Student)
- phone_number
- profile_picture

◆ Only **Instructors** can create and manage courses.

◆ Students can only view courses and enroll.

2 Authentication System

Implement:

- User Registration (using Django Forms)
- Login
- Logout
- Role-based access control
- Redirect users based on role after login:
 - Admin → Admin dashboard
 - Instructor → Instructor dashboard
 - Student → Student dashboard

3 Course Model (CRUD Operations)

Create a Course model with:

- title
- description
- price
- created_by (ForeignKey to custom user – Instructor only)
- created_at
- updated_at

4 CRUD Functionalities

Instructor Can:

- Create Course
- View Their Courses
- Update Course
- Delete Course

Student Can:

- View all courses
- Enroll in course (ManyToMany relationship)
- View enrolled courses

5 Forms

Use:

- ModelForm for:
 - User Registration
 - Course Create/Update

- Custom validation:
 - Phone number format validation
 - Price must be positive
 - Only instructors can create courses
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6 Templates

Create separate dashboards:

- Admin dashboard
- Instructor dashboard
- Student dashboard

Use Django template inheritance.

7 Extra Constraints (Important)

- Use `@login_required`
- Use `@user_passes_test` or custom decorators for role restriction
- Override `save()` method in `Course` if necessary
- Use class-based views OR function-based views
- Use messages framework for success/error notifications